

HoloFight: An Augmented Reality Fighting Game

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Figure 1: (Left) Fighter avatars interact in augmented reality on an arena in the physical environment of two of the players. (Center) Avatars can dodge explosions. (Right) When an enemy avatar gets too close, a player can pick up and move it with one hand, while still holding the controller in the other.

ABSTRACT

Augmented Reality (AR) provides opportunities to create exciting new kinds of digital entertainment, such as watching movies on a large virtual screen or playing games that interact with a real physical room. While a number of AR games have been built, many do not build on the control innovations found in modern console, PC, and mobile gaming [Von Itzstein et al. 2019]. To explore the space of immersive multiplayer experiences with support for control innovations found in common non-immersive video games, we present *HoloFight*, a multiplayer fighting game using two or more Microsoft HoloLens 2s, two or more Xbox controllers, and the various natural user interfaces supported by the Microsoft HoloLens 2.

CCS CONCEPTS

- Human-centered computing → Mixed / augmented reality;
- Applied computing → Computer games.

KEYWORDS

Gaming, augmented reality, natural user interfaces

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1 INTRODUCTION

Video games in augmented reality are gaining widespread popularity with applications such as Pokémon Go [The Pokemon Company 2021] and Harry Potter: Wizards Unite [Inc. 2021]. However, these games are restricted to mobile devices. As head worn displays (HWDs) such as the Microsoft HoloLens 2 have not yet gained widespread popularity due to price and limited consumer appeal, a lot still needs to be done to create engaging gaming experiences specifically for AR HWDs [Von Itzstein et al. 2019].

Some of the more notable games and systems for AR HWDs include Microsoft's RoboRaid [Microsoft 2021c], Microsoft's Fragments [Microsoft 2021a], and VRabl [Buckers et al. 2018]. These games take advantage of the scene understanding capabilities of the HoloLens, but some are restricted to one player and use body-centric interfaces as the only forms of input, which can introduce fatigue over long play periods, and may be unfamiliar to players accustomed to more traditional interfaces [Boring et al. 2009]. To mitigate this, our main form of input is the Xbox controller, which is designed to be used by many players for long gaming sessions. The top three video games of 2020—*Call of Duty Warzone*, *Minecraft*, and *Animal Crossing: New Horizons* [Fosbytes 2021]—all include some form of multiplayer capability. During a global pandemic, preserving a sense of connection is more important than ever. To further that sense of community, while striving for greater social presence in these experiences, we designed a multiplayer game where players can be themselves integrated into the shared digital space, whether or not they are physically colocated.

We present *HoloFight*, a multiplayer fighting game using AR HWDs, supporting a hybrid user interface (UI) (Fig. 1). Our system builds on previous work done in AR and traditional gaming by combining important aspects of both (direct character control, utilization of the physical environment, and natural UI affordances). Our system takes advantage of modern scene understanding capabilities, allows for greater immersion for players, and encourages safe remote play, especially important during the COVID-19 pandemic.

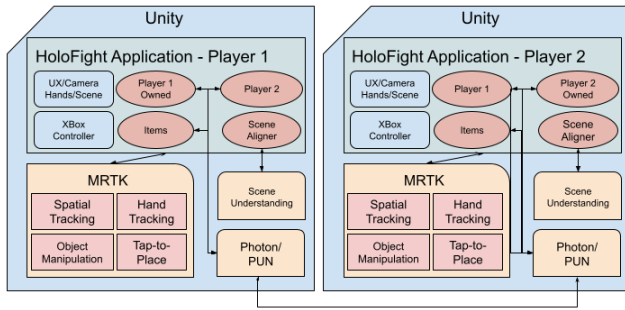


Figure 2: We build on Unity, Photon, and Microsoft’s MRTK and Scene Understanding SDK to facilitate *HoloFight*’s multiplayer and immersive capabilities.

2 HOLOFIGHT SYSTEM DESCRIPTION

HoloFight is built in Unity on top of a number of 3rd party libraries including Microsoft’s MRTK [Microsoft 2021b] and Scene Understanding APIs, Photon PUN2 Networking [Exit Games [n.d.]], Unity’s AR plugin, and our own application/game-level logic (Fig. 2). We use MRTK to track hand gestures, eye and head gaze, and the environment.

To align multiple players’ perspectives, both players perform a calibration stage before combat. Players in the same space use Azure Spatial Anchors to automatically align their perspectives. However, if they are playing remotely from different rooms, they must align their play spaces manually, using a virtual placard that automatically aligns to environment surfaces. We combine a number of traditional gaming technologies and AR interaction modalities to create the *HoloFight* experience:

- **Avatar Control.** 3D avatars move about the environment, attack, defend, and jump (damage is indicated by a health bar and animation). A player controls the avatars using the Xbox controller. To correctly model avatar movement with respect to players’ perspectives, we project the forward vector of the joystick onto the HWD’s forward direction, which is summed to the avatar’s rigid body’s current motion vector. Pressing left, up, right, and down on the joystick moves the avatar relative to the player, left, forward, right, and backward.
- **Scene Understanding and Spatial Awareness.** To provide an immersive, straightforward, and fun experience, we utilize two modes of scene reconstruction. We use the spatial awareness capability of the HoloLens 2 (describing the world through triangle meshes) to allow players to interact with objects scattered about the environment, such as players. As these meshes do not adequately smooth large flat surfaces, we also use the Microsoft Scene Understanding SDK [Microsoft 2021d], which takes the HoloLens 2 sensory information and generates basic “SceneObjects” (e.g., wall, ceiling, and floor) that are used to describe the physical world. This combination provides a more seamless experience than either approach alone.
- **Items.** Players can use their hands to grab and manipulate items that can be thrown at avatars to cause additional damage. Since players’ tracked environments can be large, we wanted to ensure that items could be found easily. Thus, items will spawn randomly in a volume whose location is determined by the average of where each player is looking. In addition to items, players can

also spawn and place new platforms around the environment, allowing avatars to reach otherwise unreachable spaces, increasing play opportunities. Both items and platforms can be controlled at a distance. This is especially useful in remote mode (described below), so players will have a greater area of shared play space.

- **Remote Mode.** In light of the COVID-19 pandemic, we realize that players need a way to engage with each other at a safe distance. Thus, players have the option to play in the same physical space or remotely. In remote mode, players can still fight each other and use items after a manual space calibration. However, they will not be able to utilize spatial understanding capabilities due to the differences in physical spaces.

3 CONCLUSIONS AND FUTURE WORK

We introduced *HoloFight*, a multiplayer augmented reality fighting game for colocated and remote players, and associated modalities, bridging traditional and immersive gameplay. Moving forward, we will merge parts of players’ environments, and expand avatar movesets to better take advantage of physical surfaces (e.g., wall jumps and runs). We will also further utilize eye tracking to create new gameplay opportunities, based on players’ gazes (e.g., opportunistic platform placement, controlling the trajectory of ballistic items). In addition, we intend to expand our remote mode, allowing players to participate through PCs or phones and conduct a user study to determine the impact of our approaches on immersion and entertainment.

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